

Multivariable Regression: Multiple Regressors

Multiple Regression With k Independent Variables

- Generally, the dependent variable Y may depend on k different independent variables. Thus in practice we are often faced with a multiple regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki} + \epsilon_i, \quad i = 1, \dots, N$$

A single coefficient β_j measures the effect of a unit change in X_j on the expected value of Y , **ceteris paribus** (holding all else constant):

$$\beta_j = \frac{\Delta \mathbb{E}(Y|X_1, \dots, X_k)}{\Delta X_j} \Bigg|_{\text{other } X\text{'s held constant}} = \frac{\partial \mathbb{E}(Y|X_1, \dots, X_k)}{\partial X_j}$$

$$\begin{aligned}\log(wage_i) = & \beta_0 + \beta_1 educ_i + \beta_2 educ_i^2 + \beta_3 exper_i + \beta_4 exper_i^2 \\ & + \beta_5 metro_i + \beta_6 south_i + \beta_7 west_i + \beta_8 midwest_i + \beta_9 female_i + \beta_{10} black_i + \epsilon_i, \quad i = 1, \dots, N\end{aligned}$$

$$\begin{aligned}
\log \widehat{(wage)}_{(se)} = & 1.5827_{(0.1838)} + 0.0357_{(0.0252)} \cdot educ + 0.0025_{(0.0009)} \cdot educ^2 \\
& + 0.0281_{(0.0040)} \cdot exper - 0.0004_{(0.0001)} \cdot exper^2 \\
& + 0.1379_{(0.0381)} \cdot metro - 0.0789_{(0.0312)} \cdot south - 0.1574_{(0.0298)} \cdot female
\end{aligned}$$